# What Is the Temperature in Provo?

CE 414 · Week 9 · in-class worksheet, done on your own

Name: _____ Date and time: _____




## 1. Gather the samples

Look up the **current air temperature** in each city (weather.com or similar). Find each city's **straight-line distance to Provo** with the scale bar or an online map's measure tool.

| City | Temp (°F) | Distance to Provo (km) |
|---|---|---|
| American Fork | | |
| Moab | | |
| Cedar City | | |

Temperature source: _____

## 2. Your own method

Estimate Provo any way you like from the three temperatures.

**Provo:** _____ °F How?

## 3. Thiessen polygons (nearest neighbor)

On the map, join each pair of known cities, draw the **perpendicular bisector** of each line, and join the bisectors into three polygons. Provo takes the value of the polygon it falls in.

**Polygon:** _____ **Provo:** _____ °F

## 4. Nearest neighbors, n = 2

Average the temperatures of the **two cities closest** to Provo.

( _____ + _____ ) ÷ 2 = **Provo:** _____ °F

## 5. Inverse distance weighted (IDW), power 2

Weight each city by **1 / d²** and take the weighted average of all three.

$$\text{Provo} = \Sigma\,(w \cdot T) / \Sigma\, w, \quad w = 1/d^2$$

| City | w = 1 / d² | w · T |
|---|---|---|
| American Fork | | |
| Moab | | |
| Cedar City | | |
| **Sum** | | |

**Provo** = Σ(w · T) ÷ Σw = _____ °F

The weights are tiny: keep three significant figures, or use 1 / (d ÷ 10)² instead. Scaling every weight by the same amount does not change the answer.

## 6. Check

**Provo's actual temperature now:** _____ °F

Which method came closest, and why?

Take a picture of this sheet and submit it on Learning Suite for today's in-class activity.